

Machine Learning Presentation

MA124 Maths by Computer

A1 · A2 · A3 · A4 · B1 · B2

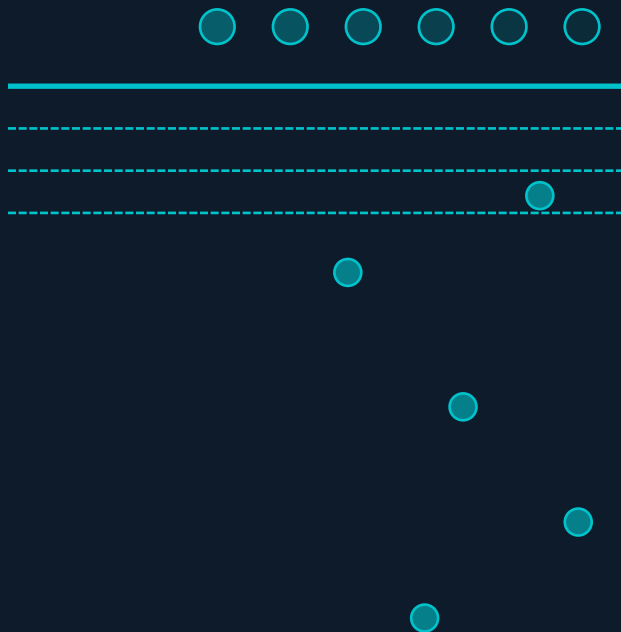


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A1

Linear Regression: Seoul Bike Dataset

Initial Explorations

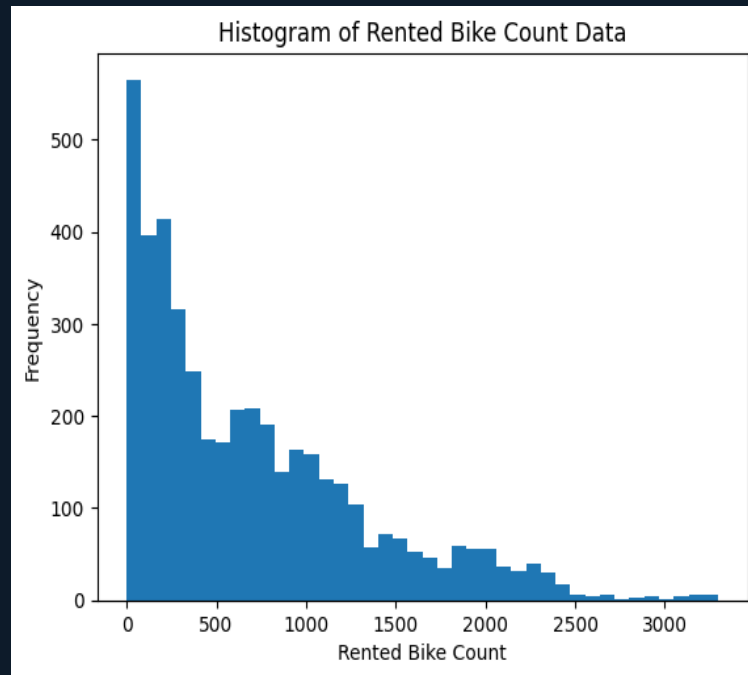
A1

Seoul Bike Rental Dataset

- Includes data on bike rentals per hour, time of day, month, and weather conditions.
- Reference paper by Sathishkumar VE, Jangwoo Park, Yongyun Cho.

Initial Findings

- Less than 500 bikes rented is most common occurrence.
- Linear regression required for further analysis.

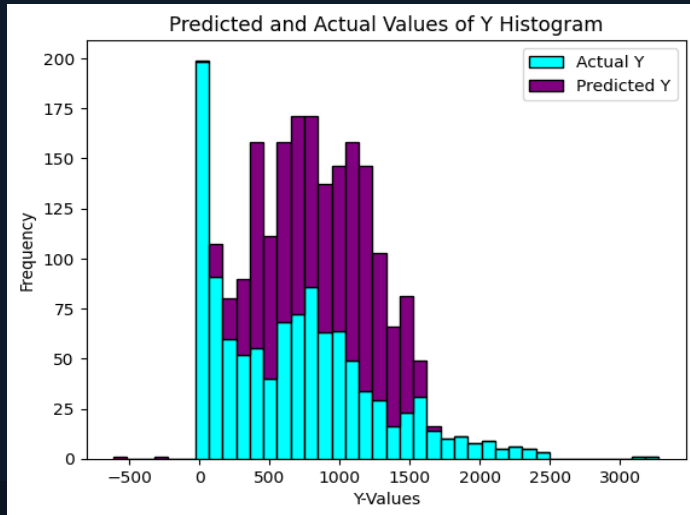


Linear Regression Model

A1

Our Model

The R2 value is 0.0936546799282616
The RMSE value is 546.1983395427221
The MAE value is 404.5364341012174
The CV value is 80.92576565129693



Reference Values

Table 4

Models performance.

Models	Training				Testing			
	R ²	RMSE	MAE	CV (%)	R ²	RMSE	MAE	CV (%)
LM	0.55	431.72	321.84	61.15	0.55	427.71	322.32	61.03

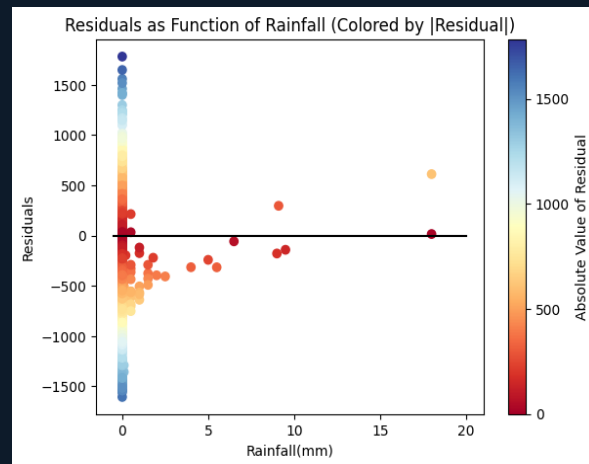
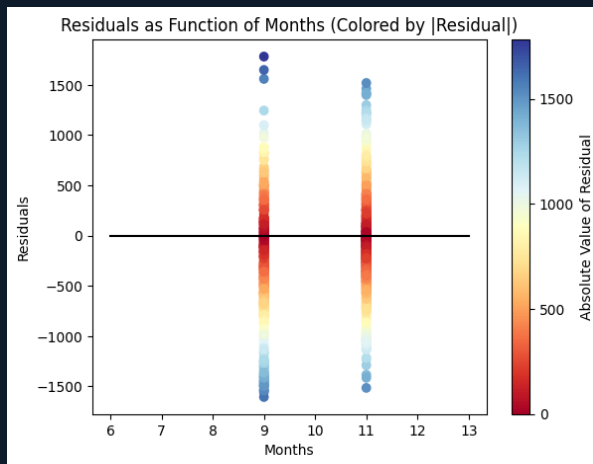
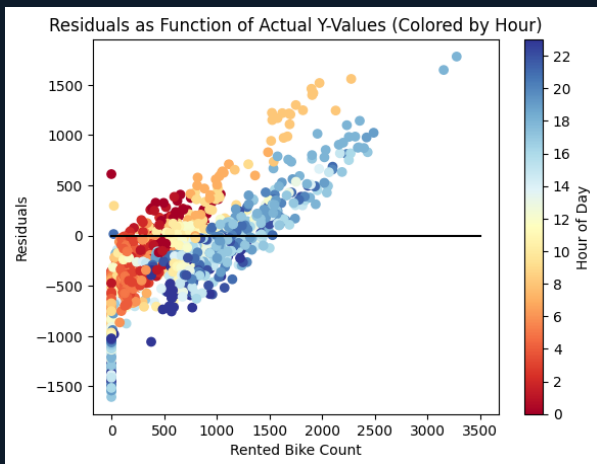
(E, S.V., Park, J., & Cho, Y., 2020)

- Relatively similar values found for:
 - Root Mean Squared Error (RMSE)
 - Mean Absolute Error (MAE)
 - Coefficient of Variation (CV)
- Modified dataset led to variation with the R² value.

Further Exploration: Residuals

A1

Residuals = difference between predicted values from model and actual values from dataset



- Positive linear correlation.
- Low levels early in day, wider range in afternoon.

- Wide spread of residuals indicates model inaccuracies.
- Months beyond 9 are test data (not included) and missing data explains other gaps.

- Large number of datapoints where no rainfall present.
- As rainfall increases, residuals cluster towards zero, indicating higher accuracy with rainfall levels.

A2

Linear Regression from Scratch

Linear Regression from Scratch on Small Dataset

A2

Task:

Design a linear regression model to find the line of best fit for 3 data points.

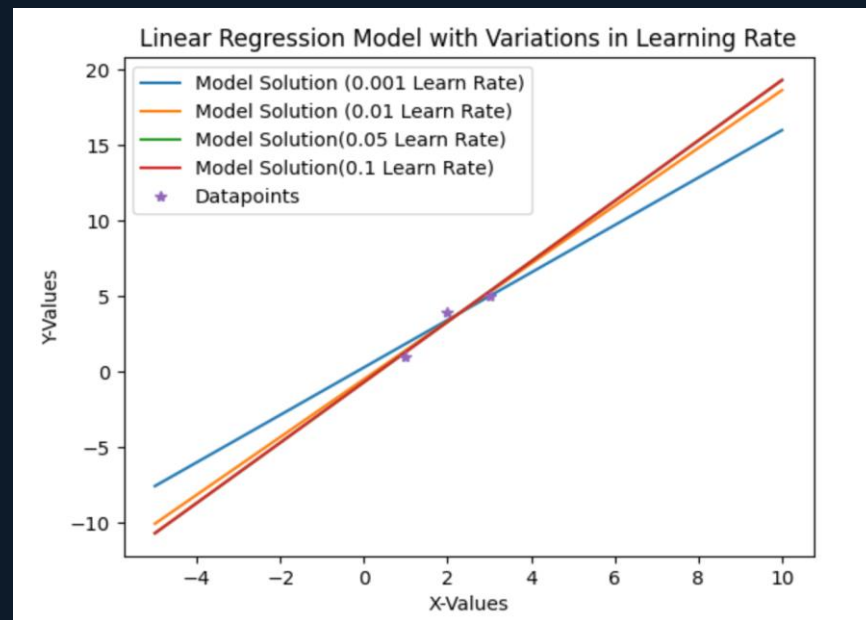
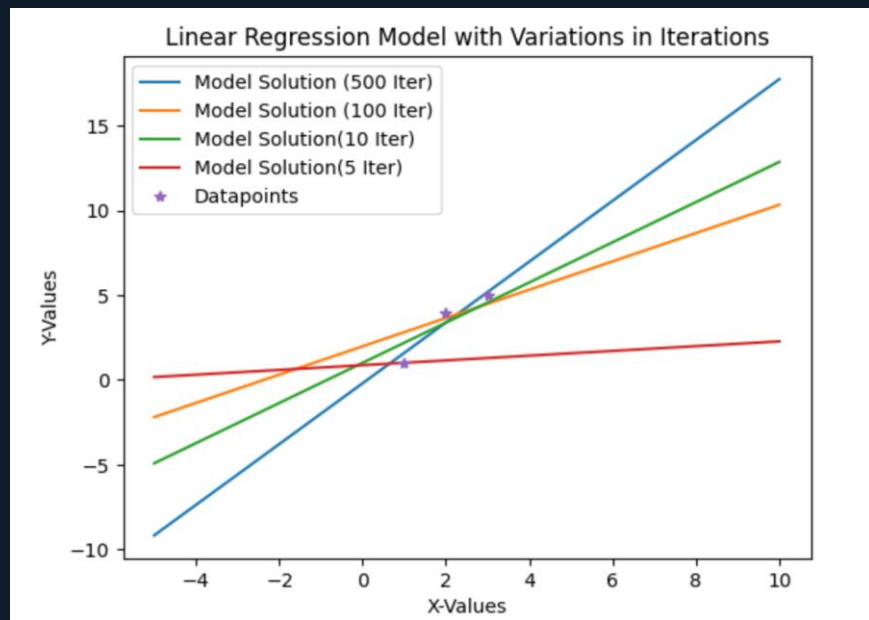
What We Did:

- Computed cost function and partial derivatives by hand.
- Found a random m and c between -2 and 2.
- Calculated the partial derivatives with respect to m and c .
- Determined new values for m and c via gradient descent.

```
def model1(alpha, num_iterations):  
    m_ = random.uniform(-2,2)  
    c = random.uniform(-2,2)  
    for i in range(0, num_iterations):  
        partial_m, partial_c = partial_derivatives(m_,c)  
        if i%(num_iterations//5) == 0:  
            print(cost(m_,c))  
        m_ = m_ - partial_m*alpha  
        c = c - partial_c*alpha  
    return(m_,c)
```


Effect of Learning Rate and Number of Iterations

A2



The number of iterations has a larger effect on the line of best fit than the learning rate.

A3

Logistic Regression: Flower Dataset

Machine Learning Binary Classification (Logistic Regression)

A3

What is it?

Logistic regression is a type of machine learning model used for **binary classification**. It predicts one of two outcomes (e.g. yes/no, red/blue).

The model looks at the input (in this case, coordinates) and **produces a probability** that a point belongs to a certain class. A threshold, usually 0.5, is then used to predict which class it belongs to.

Completing the Task

A3

What was the task?

Given a dataset of flower-shaped points labelled red or blue, build a model to predict the colour of a point based on its coordinates. Then define regions of the plane as red or blue.

How did we do it?

1 Initialise the model's parameters

2 Forward propagation

$$z = WX + b$$

$$\sigma(\lambda) = \frac{1}{1 + \exp(-\lambda)}$$

3 Compute the cost

4 Backpropagation

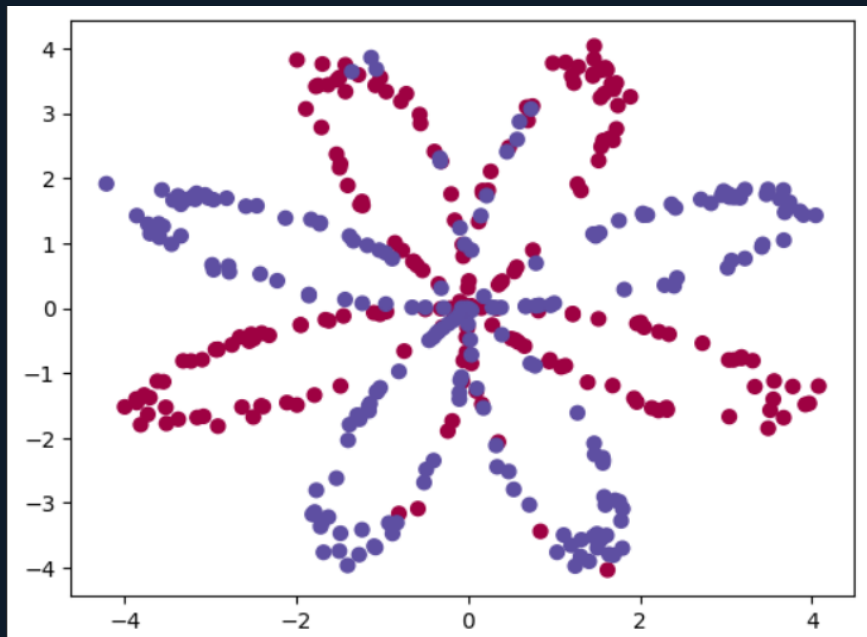
5 Update parameters

6 Predict and plot the results

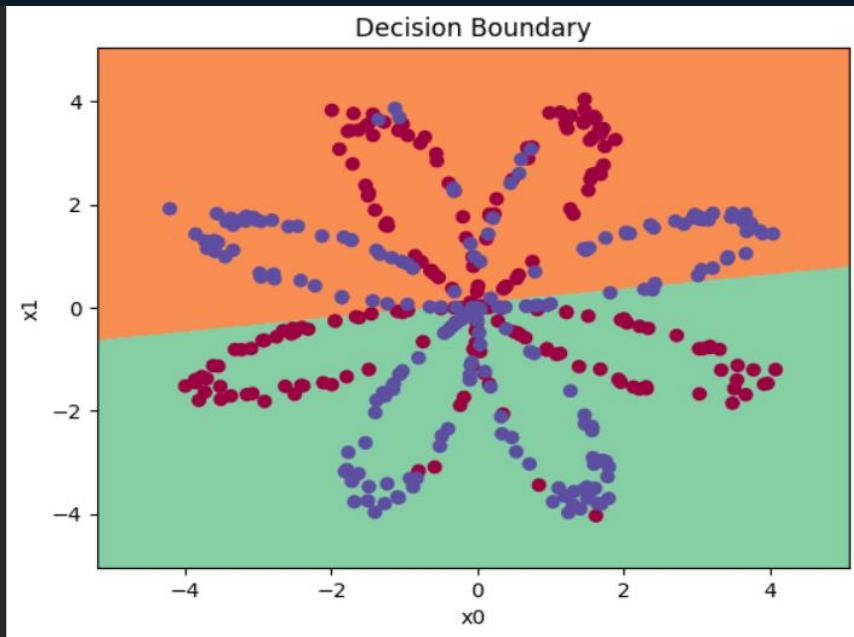
The Results

A3

Initial Data Set



Final Output



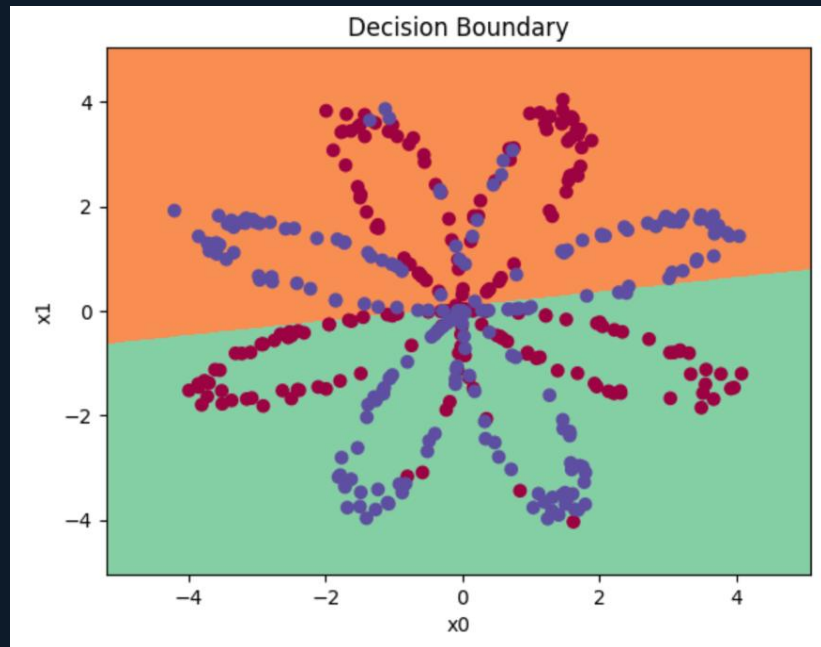
*It is clear that binary logistic regression does not work well for this data set. The data is not linearly separable, so the model performs poorly. **Accuracy: 47%.***

A4

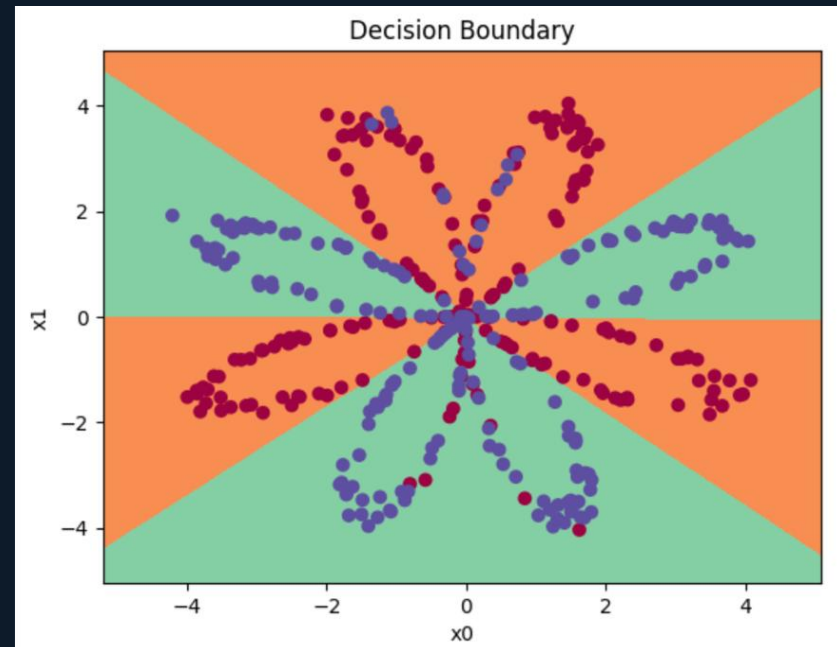
Hidden Layer Neural Network

Neural Networks: Why?

A4



47% accuracy



90% accuracy

The big conceptual difference

A4

A more complicated forward propagation step:

linear



Allows different features to be weighted (just as in A3)

tanh



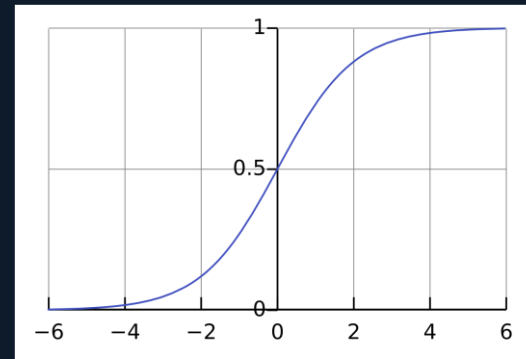
An *activation function* to introduce **non-linearity**

linear

sigmoid



A function with image (0, 1) so that the model produces probabilities



Scalar equation (A3)

$$z = \mathbf{w} \cdot \mathbf{x} + b$$

Vector equation (A4)

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

We had 4 neurons in the hidden layer for our project.

B1

Improving Neural Networks

Both **underfitting** and **overfitting** were present in our own project:

- **A3:** Logistic regression underfit the flower dataset (47% accuracy)
- **A4:** Adding a hidden layer fixed this (accuracy jumped to 90%)

Underfitting

Model is too simple; fails on both training data and new data

Overfitting

Model learns noise; great on training data; poor on new data

What is it?

Model is too simple to capture the pattern, performs badly on training AND new data.

Analogy: a student who barely revised, does badly on past papers and the real exam.

Why does it happen?

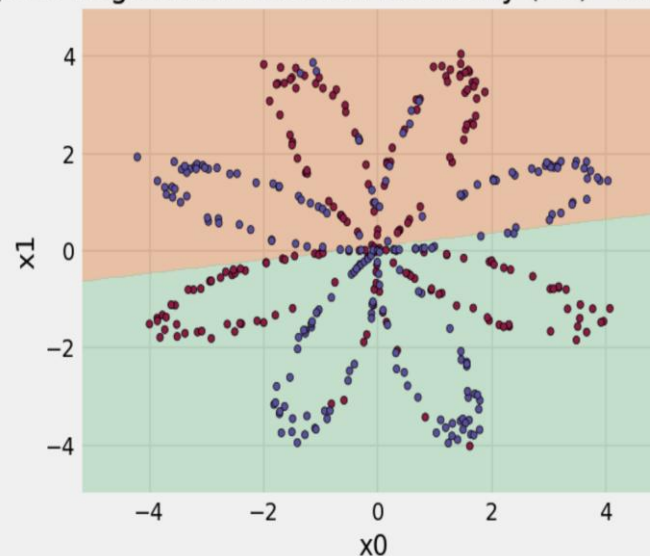
- Too few neurons or layers to represent the shape of the data
- Too few training iterations, model hasn't had time to learn
- Learning rate too high, gradient descent overshoots the minimum

How to fix it?

- Add more neurons or layers
- Train for more iterations
- Reduce the learning rate

- Logistic regression can only ever draw a straight line decision boundary
- The flower dataset is non-linear: red and blue points wrap around each other
- No straight line can separate them. 47% accuracy no matter how long we train
- This is **underfitting**: the model is fundamentally too simple for the data

Logistic Regression Decision Boundary (A3) - Underfitting



What is it?

Model learns training data too well, including the noise and random quirks.

Scores very high on training data but performs worse on new unseen data.

Analogy: student who memorises past papers word for word, aces repeats, struggles on new ones.

Why does it happen?

- Too many neurons/layers relative to the amount of training data
- Trained for too many iterations, starts memorising rather than learning

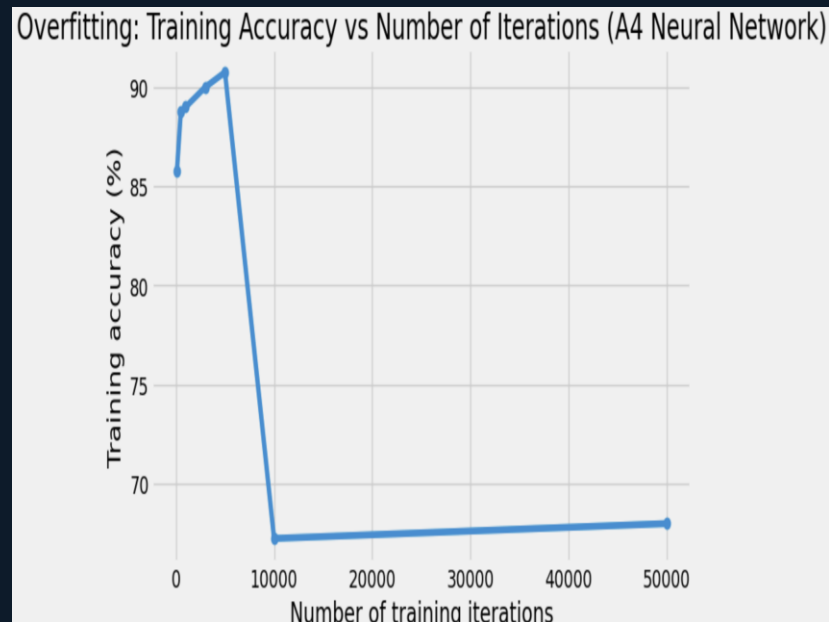
How to fix it?

- Regularization: penalises large weights (Lasso and Ridge shown in B1)
- Dropout: randomly switches off neurons during training
- Early stopping: stop when validation performance starts to drop

Link to Task A4

B1

- Trained the A4 neural network for **increasing iterations** on the flower data
- **Early on:** accuracy improves, model is genuinely learning the pattern
- **Later:** training accuracy rapidly drops, model is memorising the 400 training points
- Around 5000 iterations gives a good balance, 50,000 clearly shows overfitting
- This is why task A1 split data into train/test sets, to catch this happening



Overfitting & Underfitting: Summary

- Underfitting: model too simple, fails on both training and test data
- Overfitting: model too complex, great on training, poor on new data

Seen In Our Project

- A3 logistic regression underfit the flower dataset: 47% accuracy
- A4 neural network found the better balance: 90% accuracy

Key Fixes

- Underfitting: more neurons/layers, more iterations, lower learning rate
- Overfitting: regularization, dropout, early stopping, more data

Goal: find the sweet spot between the two

What is it?

A technique used to prevent overfitting.

Adds penalty terms and encourages more generalised and simpler models.

There are 3 types:

Lasso · Ridge · Elastic Net

For each type, the penalty term is multiplied by a constant that **controls the strength** of the regularization.

Least Absolute Shrinkage and Selection Operator (a.k.a. Lasso)

- Adds absolute value of magnitude of the coefficient as a penalty term to the loss function.
- Can shrink some coefficients to zero, this selects only the important features.
- L1 regularization.

Benefits:

- Prevents overfitting. Identifies important features and shrinks others to zero.

Cons:

- Can eliminate too many features, causing underfitting.
- If features are correlated, it arbitrarily shrinks a related coefficient to zero.

$$Cost = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{i=1}^m |w_i|$$

What is it?

- Adds square magnitude of the coefficient as the penalty term to the loss function. L2 regularization.

Benefits:

- Handles multicollinearity by shrinking coefficients of correlated features instead of eliminating them.
- Prevents overfitting.

Cons:

- Coefficients are never shrunk to zero.
- Does not perform as well if there are only a few important features.

$$Cost = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{i=1}^m w_i^2$$

What is it?

- Elastic net regression uses a mix of L1 (Lasso) and L2 (Ridge) regularization techniques.
- We add the absolute value of the coefficients and the square of the coefficients.
- An extra parameter controls the ratio of L1 and L2 regularization.
-

Using a mix of both techniques means the model handles multicollinearity and can select important features.

$$Cost = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \left(\lambda \sum_{i=1}^m |w_i| + (1 - \lambda) \sum_{i=1}^m w_i^2 \right)$$

B2

Further Machine Learning Models

Linear Regression

B2

Types of Linear Regression:

- Simple Linear Regression - one input and one output variable. Fits a straight line to predict output from trained samples.
- Multiple Linear Regression - more than one independent variable. Best-fit is a plane in multi-dimensional space.

Applications:

- **Real Estate:** predict property prices using location and size.
- **Finance:** forecast stock prices using interest rates and inflation data.
- **Agriculture:** estimate crop yield from rainfall, temperature, and soil quality.
- **E-commerce:** analyse how price, promotions and seasons affect sales.

Advantages:

Simple, computationally efficient, good baseline model, handles large datasets well.

Disadvantages:

Assumes linearity, sensitive to outliers, susceptible to overfitting and underfitting.

Simple Linear Regression

B2

This was explored in Section A. To summarise:

The relationship is modelled with a linear equation. $Y = w_0 + w_1X + \epsilon$

The main goal is to find the best-fit line that minimises the difference between actual and predicted values by estimating the parameters: w_0, w_1

Results of the example implementation on Salary Dataset (focusing on errors):

	Marks	Predicted Salary	Actual Salary	Error	Error Squared
0	34	3720.703181	3400	-320.703181	102850.530562
1	95	5044.371180	4900	-144.371180	20843.037519
2	64	4371.687443	4250	-121.687443	14807.833746
3	88	4892.474852	5000	107.525148	11561.657452
4	99	5131.169081	5100	-31.169081	971.511623
5	51	4089.594263	4600	510.405737	260514.016478

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i, y_i)^2$$

MAE: 205.97696170444905

MSE: 68591.43123004607

RMSE: 261.89965870547843

Multiple Linear Regression

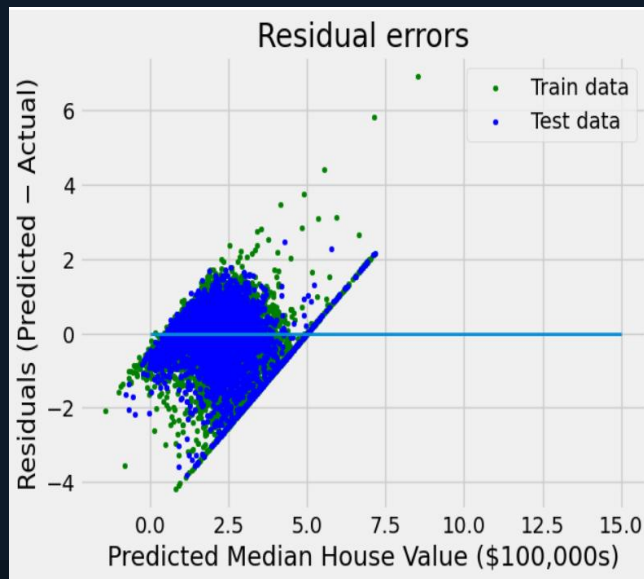
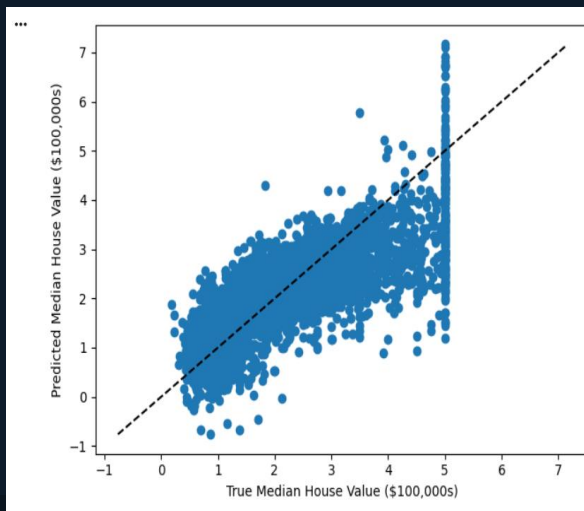
B2

How does it work?

Steps are similar to simple linear regression, but the evaluation process differs. We can find which factor has the highest influence on the predicted output and how variables relate to each other.

The linear equation is: $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$

Results:



Polynomial Regression

B2

What is it?

An extension of linear regression for complex, non-linear data. Instead of a straight line it fits curves by using polynomial features of higher degrees.

Equation form: $y = w_0 + w_1x + w_2x^2 + w_3x^3 + \dots + w_nx^n + \epsilon$

Applications:

- Agricultural yield prediction based on environmental conditions.
- Modelling electricity consumption in relation to temperature.

Advantages:

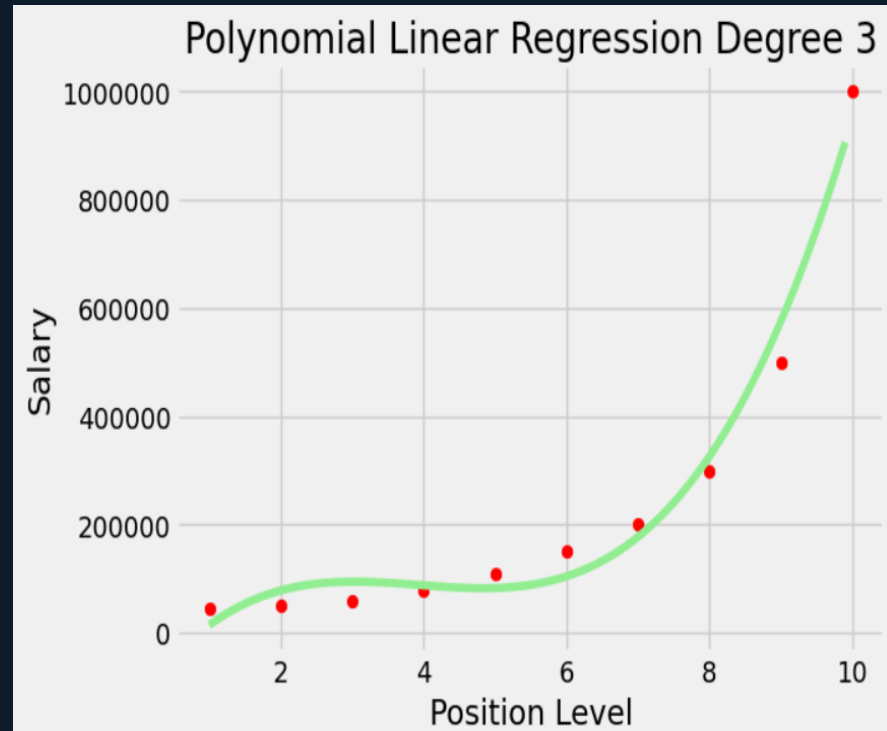
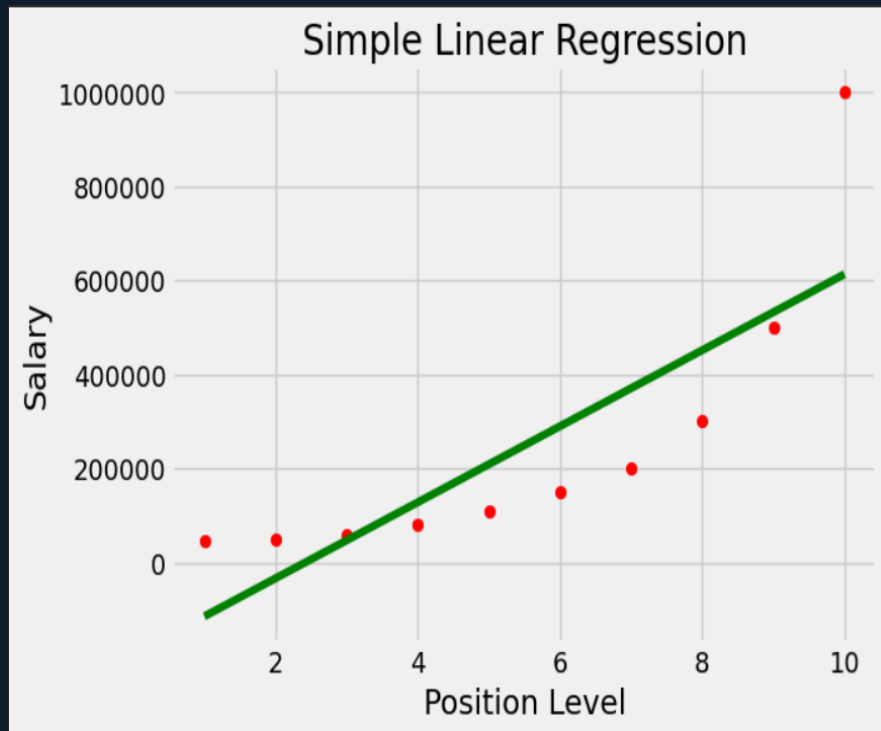
Works on any size dataset; handles non-linear problems.

Disadvantage:

Must choose the right polynomial degree for a good bias/variance trade-off.

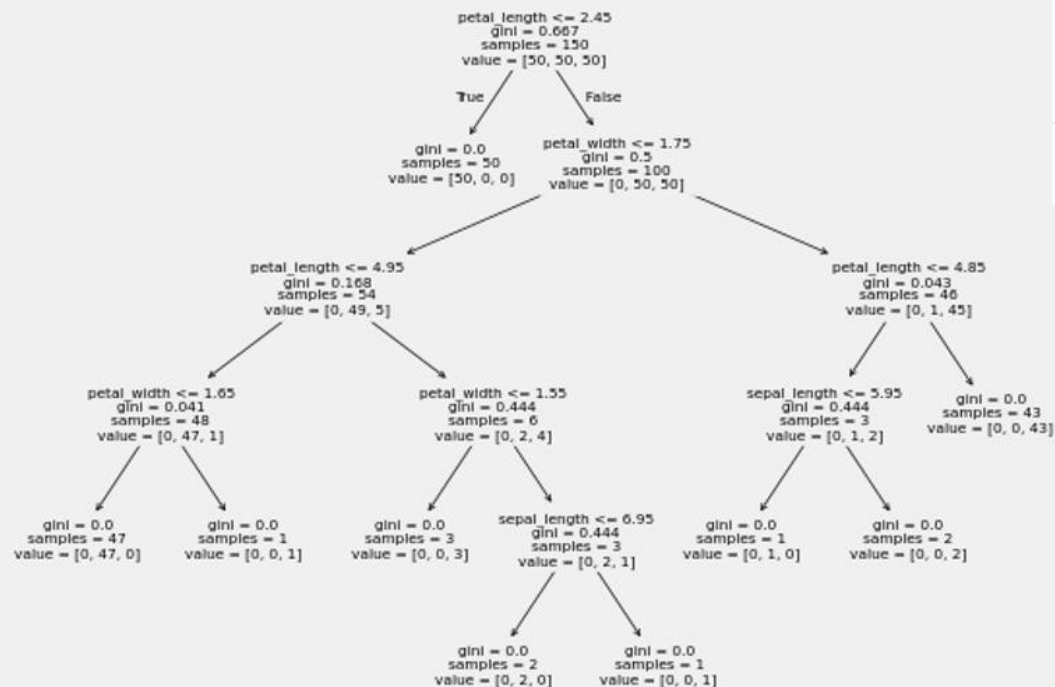
Polynomial Regression Implementation Results

B2



Decision trees

B2



Predictions

```
print(dtree.predict([[5.7, 3.8, 1.7, 0.3]]))
print(dtree.predict([[4, 3, 4, 1]]))
```

```
['Iris-setosa']
['Iris-versicolor']
```

This time we use multiple decision trees, each looking at a different part of the data, to come to a decision.

How do they decide?

- For classification tasks they take a vote.
- For regression tasks they take an average.

Classification Models (Logistic Regression)

B2

This was seen in section A. To summarise, logistic regression predicts the probability that an input belongs to a specific class.

Types of Logistic Regression:

- **Binary:** target has two outcomes (Yes/No, 0/1). Most common form, commonly used for binary classification problems.
- **Multinomial:** target has three or more nominal categories that are not ordered (e.g. classifying animal types), it extends the binary logistic regression to handle multiple classes.
- **Ordinal:** this type applies when the dependent variable has three or more categories with a natural order or ranking (e.g. low/medium/high).

Advantages:

Doesn't require high computation power, easy to implement and interpret, doesn't require feature scaling.

Disadvantages:

Cannot handle large numbers of categorical features, vulnerable to overfitting, cannot solve non-linear problems without a transformation of non-linear features.

Applications:

- Predicting likelihood of disease occurrence based on risk factors in medical research.
- Assessing probability of loan default in financial analysis.
- Determining customer churn in marketing analytics.
- Identifying sentiment polarity in text classification.

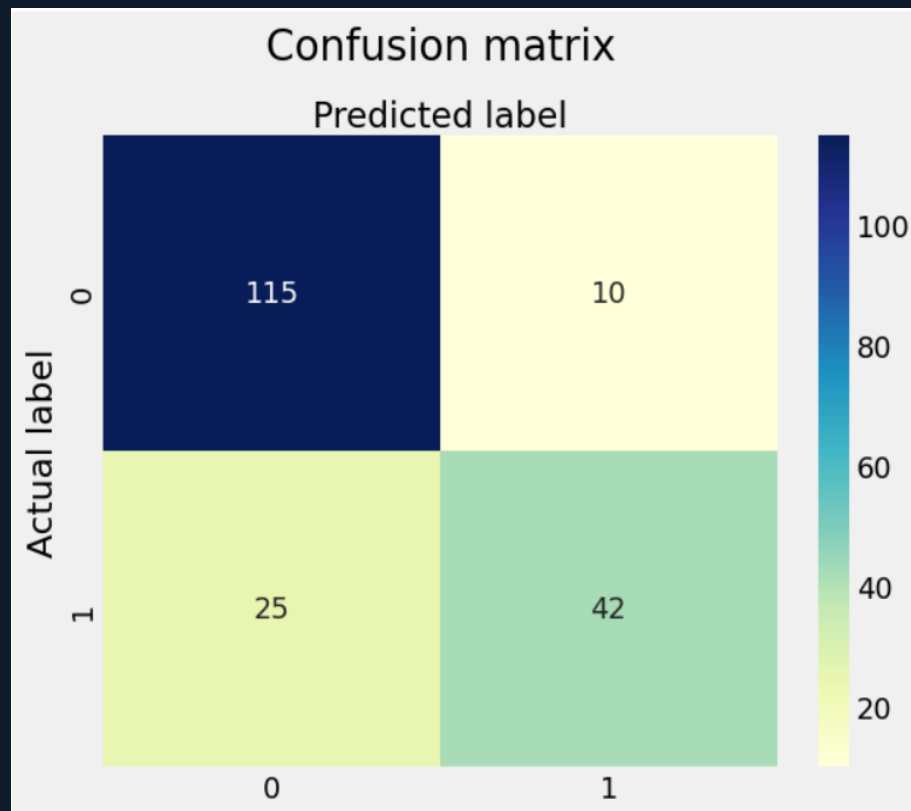
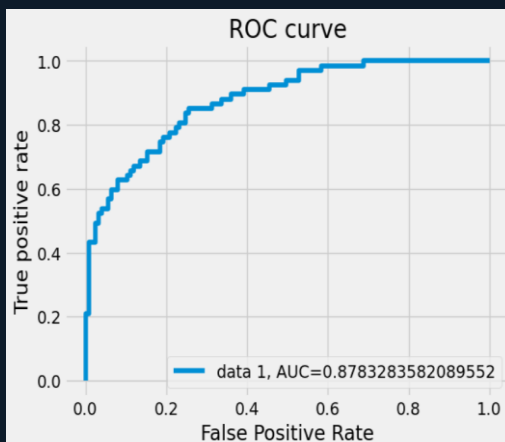
This was also explored in section A. This time, it was applied to a dataset with 7 features. We cannot plot a decision boundary (unlike A3). Instead, we look at other ways the model is evaluated.

Results of Binary Logistic Regression

B2

Classification Report:

	precision	recall	f1-score	support
without diabetes	0.82	0.92	0.87	125
with diabetes	0.81	0.63	0.71	67
accuracy			0.82	192
macro avg	0.81	0.77	0.79	192
weighted avg	0.82	0.82	0.81	192



Multinomial Logistic Regression

What is it?

Modified version of logistic regression that predicts a probability for more than two classes. Picks one category as a reference group and estimates a separate comparison for every other category against it.

The softmax function (shown below) converts raw scores into meaningful probabilities:

For a vector of scores $z = [z_1, \dots, z_k]$ where k is the number of classes:

$$\sigma(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^k (\exp(z_j))}$$

The class with the highest probability is chosen.

Example:

Predicting commuter transport choice (car/bus/train). Model produces a probability for each class, highest probability wins.

Multinomial Logistic Regression Results

B2

Applications:

- Marketing / Customer Analytics: predict which product a customer will choose.
- Medical diagnosis: predict which disease category a patient belongs to.
- Image Classification: predict which object category appears in an image.

Implementation results:

Accuracy: 96.67%

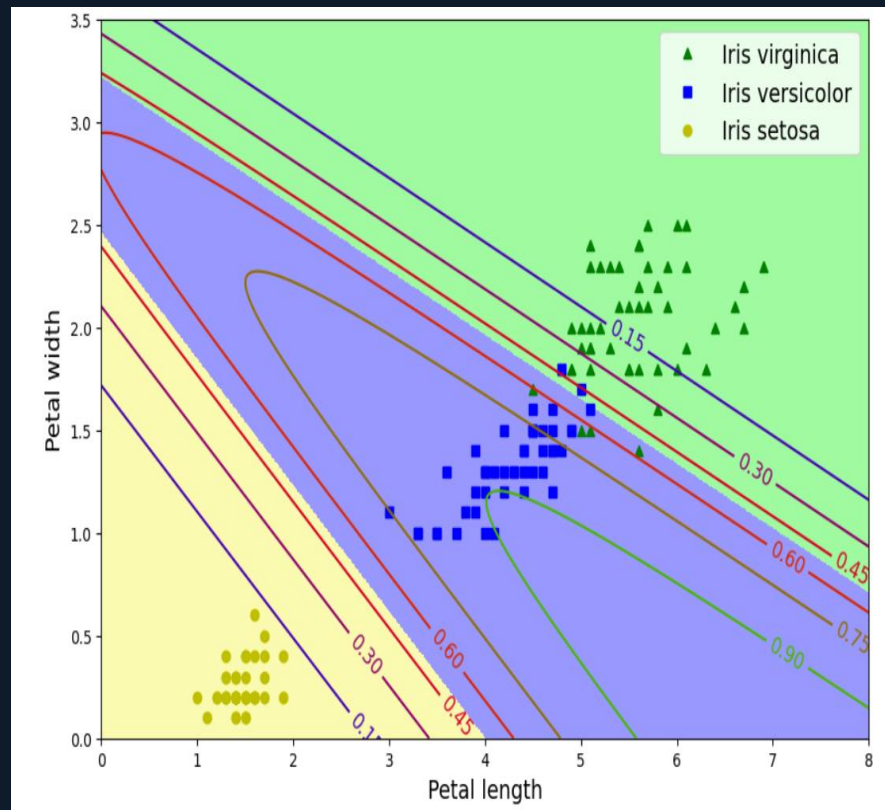
Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	0.89	1.00	0.94	8
2	1.00	0.92	0.96	12
accuracy			0.97	30
macro avg	0.96	0.97	0.97	30
weighted avg	0.97	0.97	0.97	30

class 0: 1.5887524681235827e-10

class 1: 5.090477000794684e-05

class 2: 0.9999490950711167



What is it?

- ❑ Classifies data into groups based on “close” or “similar” data.
- ❑ Supervised **instance-based** learning method.
 - ❑ Only performs computations at time of classification.
- ❑ “K” determines the **number of nearest data points** taken into account when making a decision.

Uses:

- ❑ Recommendation systems (movies, products, etc)
 - ❑ Compares user’s preferences to those of other users
- ❑ Customer segmentation
 - ❑ Groups customers by comparing shopping behaviours
- ❑ Speech recognition
 - ❑ Matches spoken words to known patterns to identify words
- ❑ Economics
 - ❑ Assessing credit-worthiness for loan applicant

Challenges

- ❑ Slow with large data (Calculates distance to every point).
- ❑ Accuracy drops with too many features.
 - ❑ Known as the “Curse of dimensionality”
- ❑ Can overfit when high-dimensional or unclean data.
 - ❑ Result of curse of dimensionality

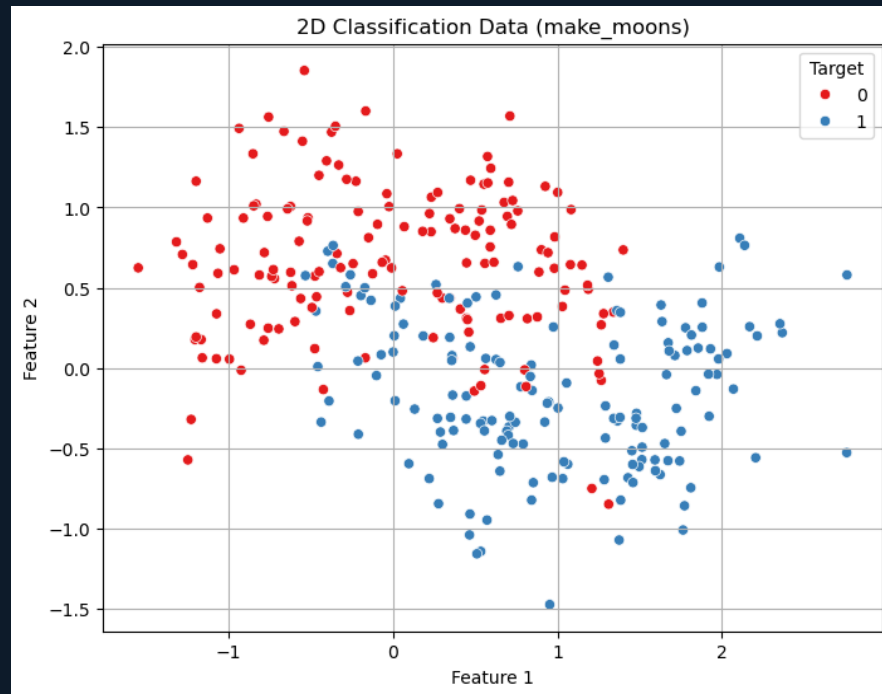
Solutions:

- ❑ Other data structures like Ball-Tree minimize computational inefficiencies.
- ❑ Feature selection and dimension reduction techniques minimize the impact of overfitting.

KNN — Method / Implementation

B2

- 1 Data generation
- 2 Split data into training and testing (70/30)
- 3 Train model
- 4 Test model
- 5 Print accuracy
- 6 Re-evaluate model's k-value by cross-validation (Best k: 7)
- 7 Rerun model with ideal k-value
- 8 Plot data points with decision boundary



KNN — Method / Implementation

B2

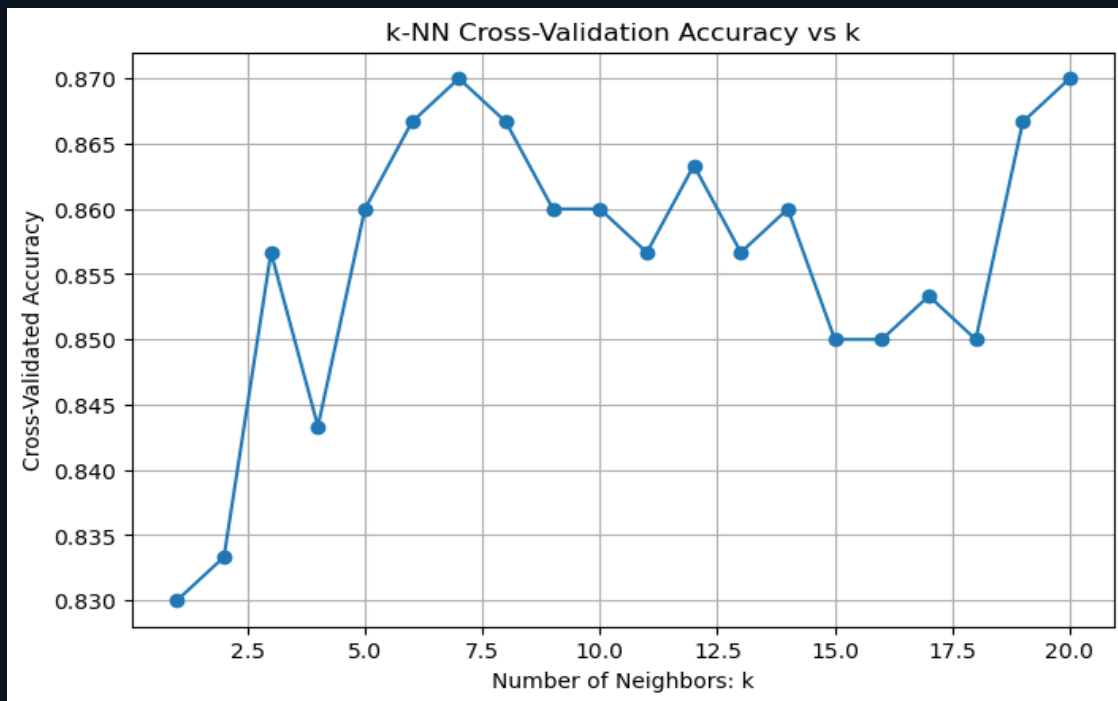
- 1 Data generation
- 2 Split data into training and testing (70/30) ←
- 3 Train model ←
- 4 Test model ←
- 5 Print accuracy ←
- 6 Re-evaluate model's k-value by cross-validation (Best k: 7)
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Test Accuracy (k=5): 0.89

KNN — Method / Implementation

B2

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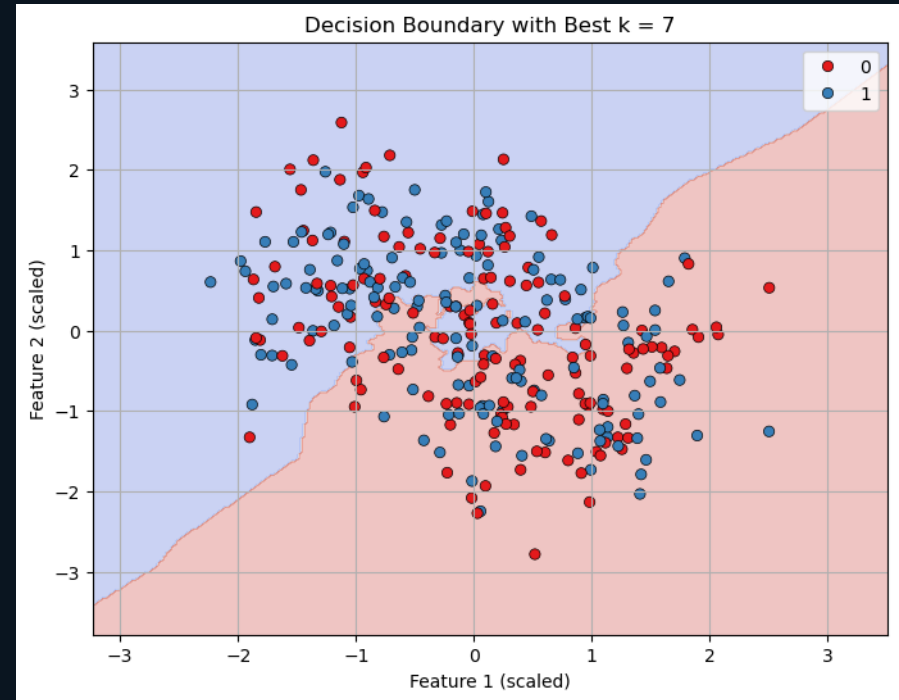


Best k from cross-validation: 7

KNN — Method / Implementation

B2

- 1 Data generation
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What is it?

- ☐ Classifies data points into **groups** based on their characteristics
- ☐ **Exclusive** clustering method
 - ☐ Data points can only belong to one cluster
- ☐ Used on **unlabeled** data points (no predefined classification structure)
- ☐ The centers of the clusters is called a **centroid**

Uses:

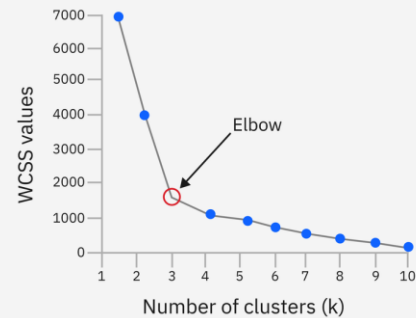
- ☐ Data segmentation
 - ☐ Ex: businesses grouping customers by shopping habits or website interaction
- ☐ Image compression
 - ☐ Pixels are clustered and averaged to reduce image size
- ☐ Anomaly detection
 - ☐ Identifying points that do not lie in clusters

Challenges:

- ❑ Choosing the number of clusters (k)

Solutions:

- ❑ Mitigated through the “Elbow Method,” which chooses k based on where the change in “within cluster sum of squares” levels off



(The “Elbow Method”, IBM, 2026)

Challenges:

- ❑ Choosing the number of clusters (k)
- ❑ Sensitive to initial centroids
 - ❑ Random initialization of centroid positions can lead to inconsistent final results.
- ❑ Non-spherical clusters
 - ❑ Automatically assumed to be spheres
- ❑ Outliers
 - ❑ Can distort centroid positions

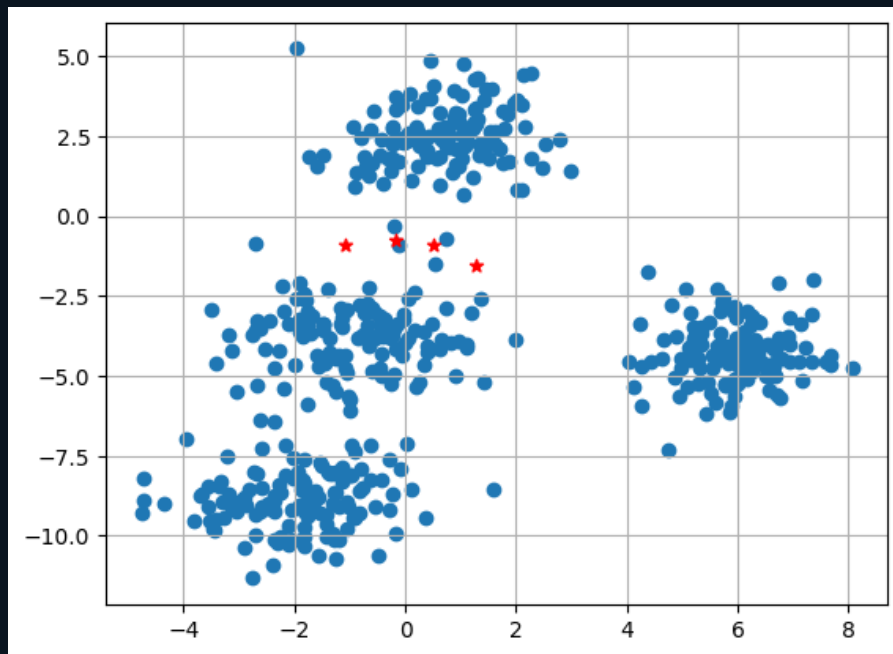
Solutions:

- ❑ Mitigated through the “Elbow Method,” which chooses k based on where the change in “within cluster sum of squares” levels off
- ❑ K-Means++ algorithm picks initial centroids recursively, which mitigates impact of randomness.

KNN — Method / Implementation

B2

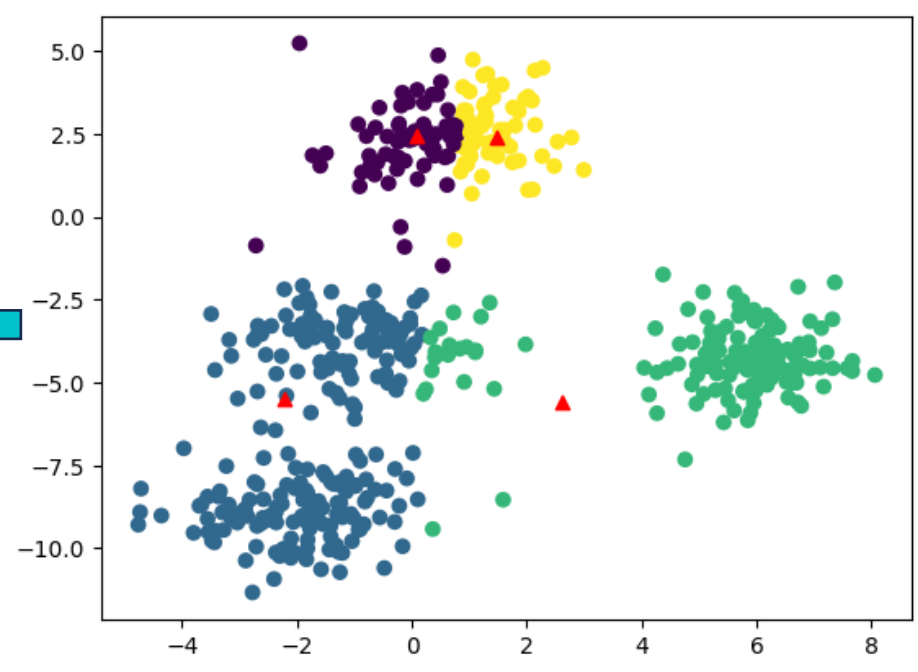
- 1 Create random dataset with 4 clusters
- 2 Initialize randomized initial centroid positions
- 3 Assign data points to nearest centroid (and its cluster)
- 4 Recalculate centroid positions by averaging points within it
- 5 Repeat until centroid positions stabilize



KNN — Method / Implementation

B2

- 1 Create random dataset with 4 clusters
- 2 Initialize randomized initial centroid positions
- 3 Assign data points to nearest centroid (and its cluster) ←
- 4 Recalculate centroid positions by averaging points within it ←
- 5 Repeat until centroid positions stabilize ←



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